

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0702

Chatom 138 kV
October 04, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0702. The Interconnection Customer proposes a 111.3 MW solar photovoltaic facility interconnecting at the Chatom 138 kV substation in the Virginia area. The request was received on November 16, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 243 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$52,393,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Tes Filer City Sta – Bridge Riv 138 kV is overloaded to 162.3% of its 206 MVA emergency rating for the loss of the Urban Substation – American 345 kV. The project contributes 8.3% DFAX (9.2 MW) to this overload.

The Detour – Cedarville Jct 138 kV is overloaded to 136.1% of its 248 MVA emergency rating for the loss of the St Louis – Nichols 138 kV. The project contributes 22.1% DFAX (24.6 MW) to this overload.

Page Substation – Purdy 138 kV line: 135.8% loading (261 MVA) following the outage of the Nichols – St Louis 138 kV; DFAX = 22.1%.

The Page – Cedarville Junction 138 kV Line is overloaded to 135.7% of its 252 MVA emergency rating for the loss of the St Louis – Nichols 138 kV line. The project contributes 22.1% DFAX (24.6 MW) to this overload.

The Nuclear Fuels – Alvin 138 kV is overloaded to 124.7% of its 201 MVA emergency rating for the loss of the Sans Souci – Oak Creek Sta 138 kV L. The project contributes 18.4% DFAX (20.5 MW) to this overload.

The Chatom – Alvin 138 kV is overloaded to 123.6% of its 208 MVA emergency rating for the loss of the Sans Souci – Oak Creek Sta 138 kV. The project contributes 18.4% DFAX (20.5 MW) to this overload.

For the loss of Santee Cooper Substation – Greenland Gap 345 kV, the Crozet – Rvital 138 kV loads to 118.7% of its 167 MVA rating. Project DFAX: 5.1%.

The Siloam Springs – Detour 138 kV Line is overloaded to 113.9% of its 181 MVA emergency rating for the loss of the Detour – Ft Benton 138 kV Line. The project contributes 14.7% DFAX (16.4 MW) to this overload.

For the loss of Midset – Brooklyn City N Plant 138 kV line, the Abotsford – Bridge River 138 kV line loads to 110.9% of its 210 MVA rating. Project DFAX: 6.9%.

Brooklyn City N Plant – Midset 138 kV: 110.4% loading (227 MVA) following the outage of the Abbotsford – Bridge Riv 138 kV; DFAX = 6.4%.

Mount Jackson – Cherry 138 kV L: 109.1% loading (178 MVA) following the outage of the Chatom – Purdy Substation 138 kV; DFAX = 7.9%.

Siloam Springs – Lewisburg 138 kV: 107.2% loading (246 MVA) following the outage of the Detour – Ft Benton 138 kV; DFAX = 6.3%.

The Cherry – Dpc Prairie Du Chien 138 kV line is overloaded to 105.9% of its 192 MVA emergency rating for the loss of the Purdy – Catom 138 kV. The project contributes 7.9% DFAX (8.7 MW) to this overload.

For the loss of Santee Cooper – Greenland Gap 345 kV L, the Dale Mabry Substation – Revital 138 kV L loads to 105.4% of its 193 MVA rating. Project DFAX: 5.1%.

Detour – Ft Benton 138 kV line: 104.8% loading (250 MVA) following the outage of the Detour – Siloam Springs 138 kV; DFAX = 15.9%.

Dale Mabry – Grottoes 138 kV: 102.3% loading (215 MVA) following the outage of the Greenland Gap Substation – Santee Cooper 345 kV Line; DFAX = 5.1%.

The Roosevelt Rd – Elkton D.P 138 kV is overloaded to 101.4% of its 197 MVA emergency rating for the loss of the St Louis – Nuclear Fuels Substation 138 kV. The project contributes 12.6% DFAX (14.0 MW) to this overload.

For the loss of St Louis – Nuclear Fuels 138 kV line, the Houston – Elkton D.P 138 kV loads to 100.4% of its 199 MVA rating. Project DFAX: 12.6%.

Wadsworth – Roosevelt Rd 138 kV: 100.1% loading (208 MVA) following the outage of the St Louis – Nuclear Fuels 138 kV; DFAX = 12.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bridge Riv – Tes Filer City Sta 138 kV	Rebuild line with higher-capacity conductor	\$5,783,000	47% (\$2,707,000)
Cedarville Jct – Detour 138 kV	Rebuild line with higher-capacity conductor	\$12,184,000	49% (\$5,956,000)
Purdy – Page 138 kV	Rebuild line with higher-capacity conductor	\$2,793,000	49% (\$1,365,000)
Cedarville Junction – Page 138 kV	Rebuild line with higher-capacity conductor	\$8,546,000	49% (\$4,177,000)
Nuclear Fuels – Alvin Substation 138 kV Line	Rebuild line with higher-capacity conductor	\$6,052,000	69% (\$4,165,000)
Chatom – Alvin 138 kV	Rebuild line with higher-capacity conductor	\$4,992,000	69% (\$3,435,000)
Crozet – Revital 138 kV	Rebuild line with higher-capacity conductor	\$5,840,000	12% (\$681,000)
Siloam Springs – Detour 138 kV Line	Rebuild line with higher-capacity conductor	\$14,433,000	100% (\$14,433,000)
Abotsford – Bridge River 138 kV line	Rebuild line with higher-capacity conductor	\$7,381,000	40% (\$2,968,000)
Midset – Brooklyn City N Plant 138 kV	Rebuild line with higher-capacity conductor	\$8,364,000	47% (\$3,911,000)
Cerry – Mount Jackson 138 kV	Reconductor line	\$1,243,000	100% (\$1,243,000)
Siloam Springs – Lewisburg 138 kV L	Reconductor line	\$1,819,000	51% (\$933,000)
Cherry – Dpc Prairie Du Chien 138 kV line	Reconductor line	\$1,759,000	100% (\$1,759,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dale Mabry Substation – Revital 138 kV L	Reconductor line	\$2,793,000	12% (\$326,000)
Detour – Ft Benton 138 kV line	Reconductor line	\$3,675,000	72% (\$2,663,000)
Dale Mabry – Grottoes 138 kV	Reconductor line	\$1,493,000	12% (\$174,000)
Roosevelt Rd – Elkton D.P 138 kV	Reconductor line	\$458,000	45% (\$204,000)
Houston – Elkton D.P 138 kV	Reconductor line	\$865,000	45% (\$385,000)
Roosevelt Rd – Wadsworth 138 kV	Reconductor line	\$2,038,000	45% (\$908,000)
	Total allocated to project		\$52,393,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.